

- 5 Cathelicidin LL-37 is a cell signalling compound that stimulates many different cells in humans. One role of cathelicidin LL-37 is stimulating the production of endothelial cells in the formation of capillaries during wound healing.

- (a) (i) Explain how it is possible for many different cell types to respond to the same cell signalling compound.

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..... [2]

- (ii) Describe the appearance of the endothelial cells of a capillary.

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..... [2]

Cathelicidin LL-37 is a protein composed of 37 amino acids.

Table 5.1 shows:

- the sequence of the first 10 amino acids in the primary structure of cathelicidin LL-37
- DNA triplets in the non-transcribed strand in the gene that codes for the first 10 amino acids in the primary structure of cathelicidin LL-37.

Table 5.1

amino acid position	1	2	3	4	5	6	7	8	9	10
amino acid	leu	leu	gly	asp	phe	phe	arg	lys	ser	lys
DNA triplet	CTG	CTG	GGT	GAT	TTC	TTC	CGG	AAA	TCT	AAA

Table 5.2 shows the triplets of bases in **DNA** and the amino acids for which they code.

Table 5.2

		second base									
		T		C		A		G			
first base	T	TTT	phe	TCT	ser	TAT	tyr	TGT	cys	T	third base
		TTC		TCC		TAC		TGC		C	
		TTA	leu	TCA		TAA	stop	TGA	stop	A	
		TTG		TCG		TAG		TGG	try	G	
	C	CTT	leu	CCT	pro	CAT	his	CGT	arg	T	
		CTC		CCC		CAC		CGC		C	
		CTA		CCA		CAA	gln	CGA		A	
		CTG		CCG		CAG		CGG		G	
	A	ATT	ile	ACT	thr	AAT	asp	AGT	ser	T	
		ATC		ACC		AAC		AGC		C	
		ATA	ile	ACA		AAA	lys	AGA	arg	A	
		ATG	met	ACG		AAG		AGG		G	
	G	GTT	val	GCT	ala	GAT	asp	GGT	gly	T	
		GTC		GCC		GAC		GGC		C	
		GTA		GCA		GAA	glu	GGA		A	
		GTG		GCG		GAG		GGG		G	

(b) Mutations of DNA base sequences in a gene can affect the primary structure of proteins.

Use the information in Table 5.1 and Table 5.2 to suggest the effect on the primary structure of cathelicidin LL-37 of:

(i) the substitution of the base T with the base A in the middle of the triplet at position 5

..... [1]

(ii) the deletion of base T in the triplet at position 2

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 [2]

(iii) the insertion of base G between bases G and T in the triplet at position 3.

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 [2]

- (c) The genetic code is described as universal.

Explain why the genetic code is described as universal.

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- (d) Use Table 5.2 to explain why some mutations have no effect on the primary structure of a protein.

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..... [2]

[Total: 12]